

REHAN FIROZ

📞 +91-8887153215 ✉ rehanfiroz0786@gmail.com  linkedin.com/in/rehan-firoz-12  github.com/rehan-firoz

Education

Lovely Professional University

M.Tech in Computer Science & Engineering (AI/ML)

Oct 2025 – Present

SGPA: 9.33/10

Babu Banarsi Das Engineering College

B.Tech in Computer Science & Engineering

Jul 2020 – Aug 2024

SGPA: 8.01/10

Sant Atulanand Convent School

Intermediate (10+2)

May 2018 – Apr 2019

83.67%

Projects

Occlusion-Robust Gait Recognition | *PyTorch, CNN, Streamlit, CASIA-B*

[GitHub](#) | Jan 2026

- Designed **OcclusionGaitNet** — a CNN with 4-part horizontal pooling, SE-style part attention, and a learnable occlusion reliability estimator that down-weights corrupted body regions during feature fusion.
- Trained on **CASIA-B** (124 subjects, 11 views) with 4-type synthetic occlusion augmentation; achieved **15.1% Rank-1** accuracy on occluded sequences vs **2.6% random baseline** — a **5.8× improvement**.
- Implemented batch-hard triplet loss + center loss + occlusion consistency regularization; resolved a `sqrt(0)` gradient instability in pairwise distance computation via distance clamping to **1e-12**.
- Deployed interactive **Streamlit** demo with per-part reliability visualization, top-3 cosine similarity matching, and runtime weight fetching via **gdown**; associated IEEE paper under preparation.

Fake News Detection System | *Python, BERT, BiLSTM, Streamlit*

[Github](#) | Sep 2025

- Built a hybrid **BERT + BiLSTM** model achieving **>98% accuracy** on synthetic test cases and **93%+ specificity** on real political news.
- Constructed a combined dataset from **WELFake, LIAR, and Synthetic AI claims** to improve generalization and prevent catastrophic forgetting.
- Conducted EDA to identify linguistic differences, balance datasets, and align preprocessing pipelines across heterogeneous sources.
- Deployed a **Streamlit** application with real-time detection, confidence scoring, and interactive visualization tools.

Customer Relationship Management System | *Java, XAMPP, SQL*

[Github](#) | Sep 2023

- Developed CRM software integrating data from multiple departments to provide a holistic, real-time view of each customer.
- Designed to enable online store owners to manage sales and core business operations efficiently over the internet.
- Implemented robust **data management** and UI components using **Java** and **SQL**.

Technical Skills

Languages: C/C++, Java, Python, HTML, CSS

AI/ML: Machine Learning, Deep Learning, NLP, CNN/RNN/LSTM, Transformer Models (BERT, GPT), Data Preprocessing, Model Evaluation, Hyperparameter Tuning

Developer Tools: VSCode, GitHub, Eclipse, Atom, Sublime

Professional Skills: Object-Oriented Programming (OOP), Front-End Web Development

Core Courses: Algorithms & Data Structures, Discrete Mathematics, Operating Systems, Networking, Digital Design

Honours and Awards

IEEE Conference Paper (Accepted)

Jan 2025

AI-Driven NPC Intelligence & Procedural Content Generation

- Accepted IEEE conference paper on AI-driven NPC behavior and procedural content generation; presentation scheduled.

Academic Excellence

1st Division with Distinction — B.Tech

- Graduated with **1st Division with Distinction** in B.Tech CSE; currently maintaining **9.33 SGPA** in M.Tech AI/ML.